

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Regular/ Supplementary Winter Examination – 2024					
Course: B. Tech		Branch : Computer Engineering/Computer Science and Engineering			
Subject Code & Name: BTCOC502, Theory Of Computation			Semester: V		
Max Marks: 60		Date: 08/02/2025		Duration: 3 Hr.	
Instructions to the Students:					
1. Each question carries 12 marks.					
2. Question No. 1 will be compulsory and include objective-type questions.					
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.					
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.					
5. Use of non-programmable scientific calculators is allowed.					
6. Assume suitable data wherever necessary and mention it clearly.					
				(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)				12
1	Which of the following is not a part of 5-tuple finite automata?			Understand	1
	a. Input alphabet	b. Transition function	c. Initial State		
2	RR* can be expressed in which of the forms:			Analyze	1
	a. R ⁺	b. R ⁺ ∪ R	c. R ⁺ ∩ R		
3	The number of states required by an automaton to accept string 101 are			Analyze	1
	a. 3	b. 4	c. 5		
4	Regular expression for string that start with ab and end with ba is			Design	1
	a. aba*b*ba	b. ab(ab)*ba	c. ab(a+b)*ba		
5	Regular expressions are closed under			Understand	1
	a. Union	b. Concatenation	c. Kleene Star		
6	Which grammar is in Greibach normal form (GNF) ?			Understand	1
	a. S → AA 0 , A → SS 1	b. A → aB B → b	c. A → aBd B → b		
7	Moore machine produces output over the change of			Understand	1
	a. transitions	b. state	c. input symbol		
8	Can Deterministic Finite automata determine palindrome number ?			Analyze	1
	a. yes	b. no	c. yes for some cases		

9	A grammar that produces more than one parse tree is called				Understand	1	
	a. regular	b. ambiguous	c. unambiguous	d. linear			
10	The language accepted by Pushdown automata is				Understand	1	
	a. type 0	b. type 1	c. type 2	d. type 3			
11	The language is recognized by Turing machine if it is generated by _____ grammar				Understand	1	
	a. type 0	b. type 1	c. type 2	d. type 3			
12	The Pushdown automata behaves like Turing machine if it has auxiliary memory				Understand	1	
	a. 0	b. 2	c. 2 or more	d. none of these			
Q.2 Solve the following.							12
A)	State some applications of regular expressions . Write regular expressions for the following (i) one 'a' followed by some number of b's . (ii) some positive number of a's followed by exactly one b . (iii) start and end with a and in between only some b's.				Create	6	
B)	Design deterministic finite automaton which consists of strings of 0's and 1's but ends with string 00, over alphabet $\Sigma = \{0,1\}$.				Design	6	
Q.3 Solve the following.							12
A)	Construct an epsilon non deterministic finite automata (ϵ - NFA) accept either a or abb or a^*b^+ .				Create	6	
B)	Distinguish between Moore machine and Mealy machine .				Understand	6	
Q.4 Solve Any Two of the following.							12
A)	Let the grammar consists of following productions . $S \rightarrow aAS \mid a$ $A \rightarrow SbA \mid SS \mid ba$ Find the leftmost and rightmost derivations for string aabbaa. Also draw the parse trees.				Apply	6	
B)	Explain Chomsky classification of grammar.				Understand	6	
C)	Construct finite automaton for the following grammar. $S \rightarrow aA \mid \epsilon$, $A \rightarrow aA \mid bB \mid \epsilon$, $B \rightarrow bB \mid \epsilon$					6	
Q.5 Solve Any Two of the following.							12
A)	Find the reduced grammar equivalent to following context free grammar where $V=(A,B,C,S)$ $T=(a,b)$ and $P = (S \rightarrow AB \mid CA ,$				Apply	6	

	$B \rightarrow BC AB, A \rightarrow a, C \rightarrow aB b$).		
B)	Explain the components of Pushdown automata.	Understand	6
C)	Explain with example how Pushdown automata is represented using transition diagram and transition table .	Understand	6
Q-6	Solve Any Two of the following.		12
A)	Distinguish between tape and head of Turing machine and Pushdown automata.	Understand	6
B)	Design Turing machine to accept the language of 0's and 1's ending with 00.	Design	6
C)	Give an example of Turing machine which has an infinite loop and never halts .	Analyze	6
	*** End ***		

51645162

51645162

51645162

51645162

51645162

51645162